Practical-7

Input-

keathana@Chandu: ~/practical-7

GNU nano 8.7.1 memory.c *

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 10;

void code_function()
{
    printf("Address of code function : %p\n", (void *)code_function);
}

int main()
{
    static int static_var = 20;
    int stack_var = 30;

    int *heap_var = (int *)malloc(sizeof(int));
    *heap_var = 40;

    printf("========================================\n");
    printf("       PRACTICAL 7 - MEMORY LAYOUT\n");
    printf("========================================\n");

    printf("Process ID : %d\n", getpid());

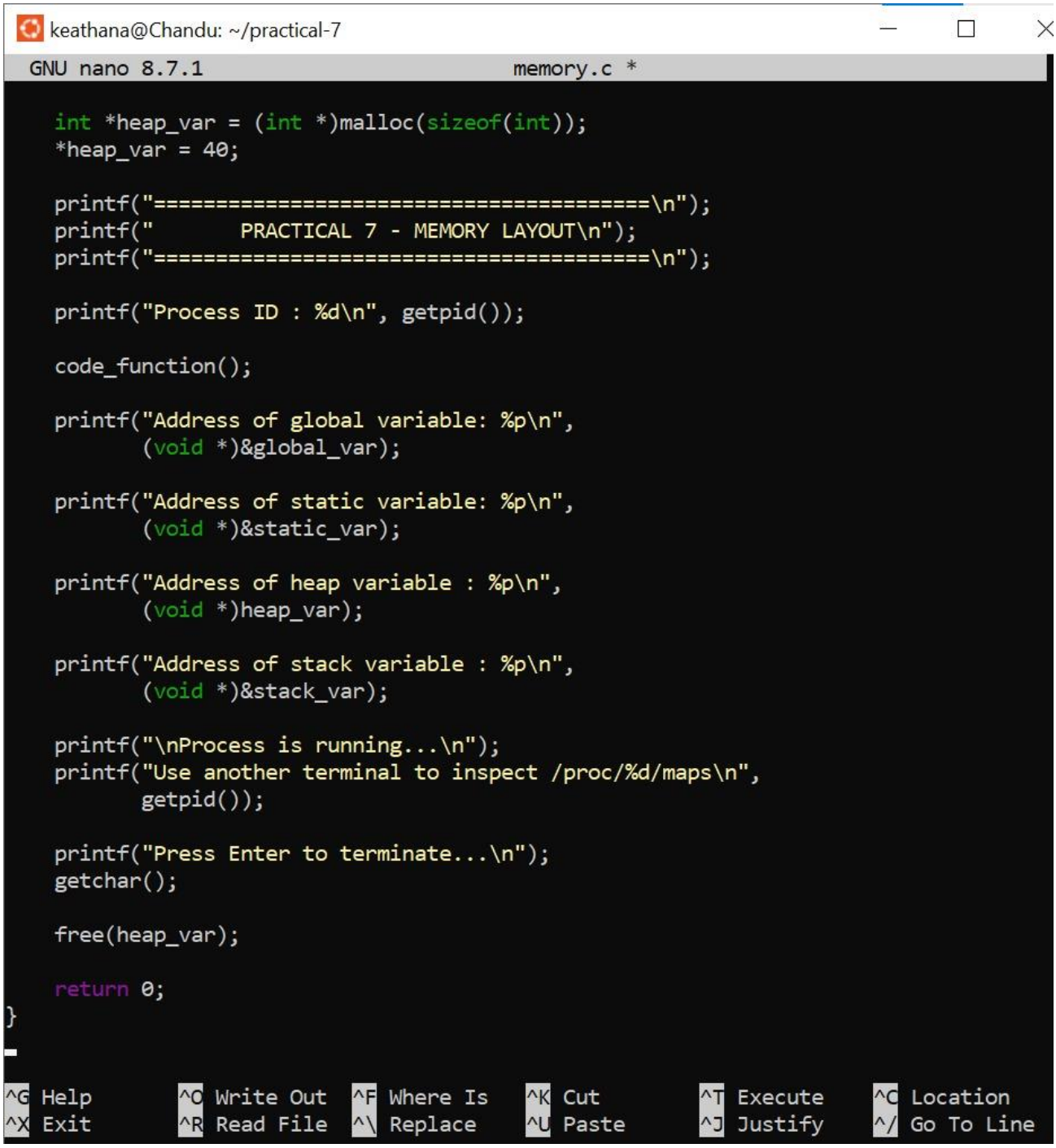
    code_function();

    printf("Address of global variable: %p\n",
           (void *)&global_var);

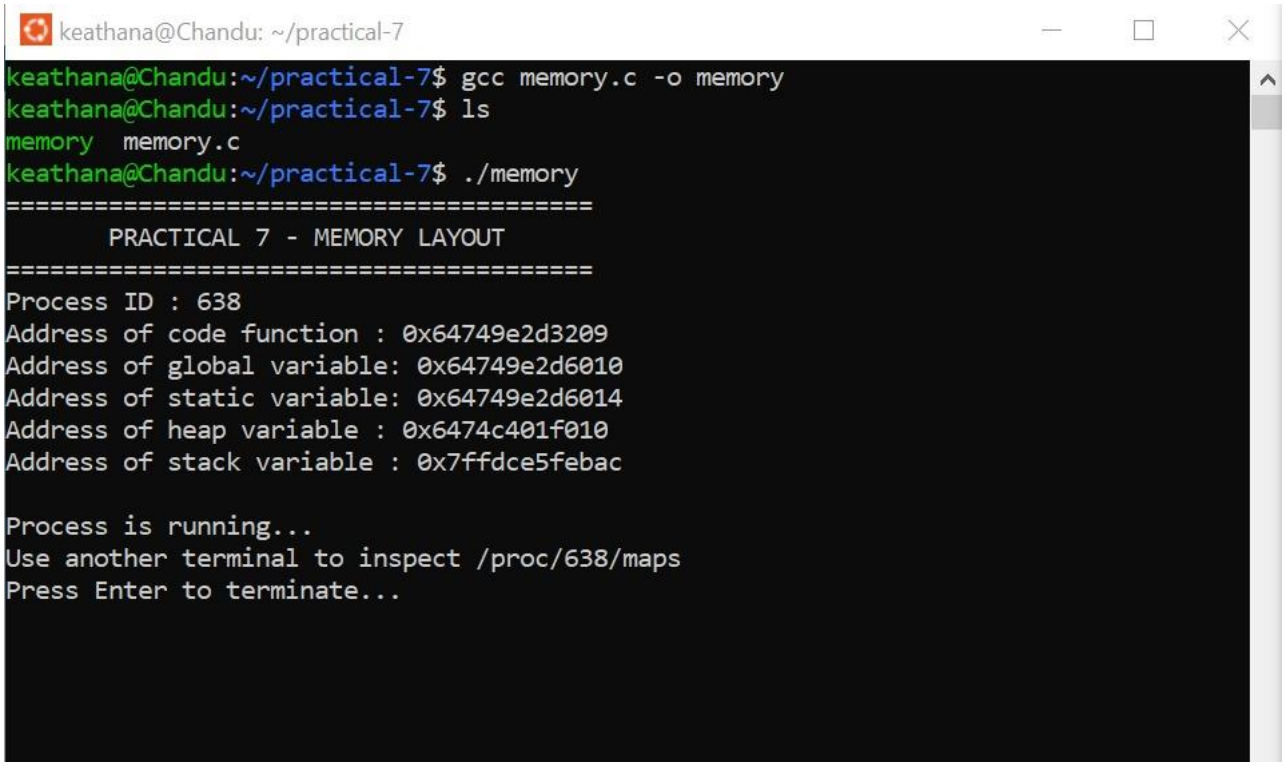
    printf("Address of static variable: %p\n",
           (void *)&static_var);

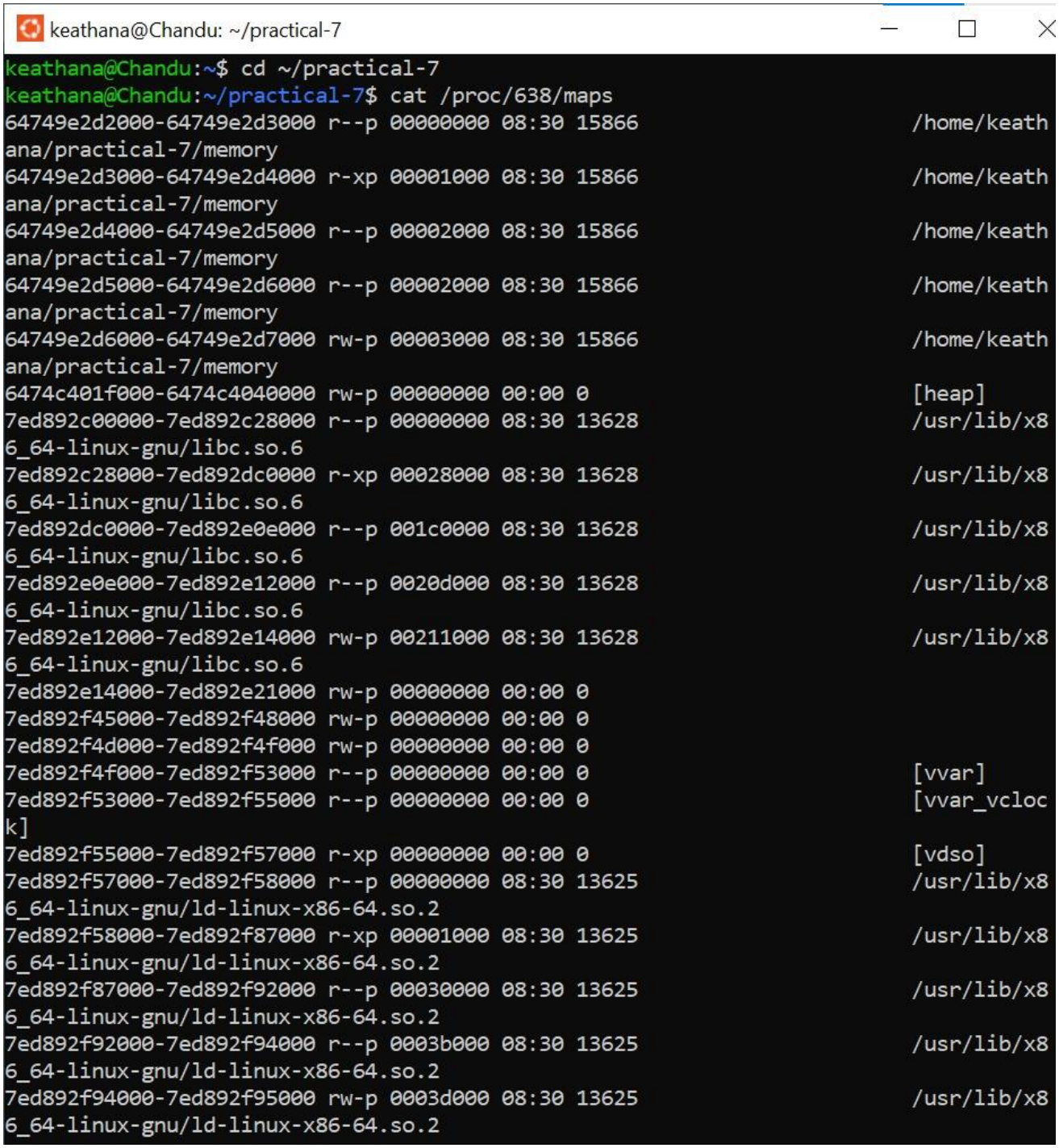
    printf("Address of heap variable : %p\n",
           (void *)heap_var);
```

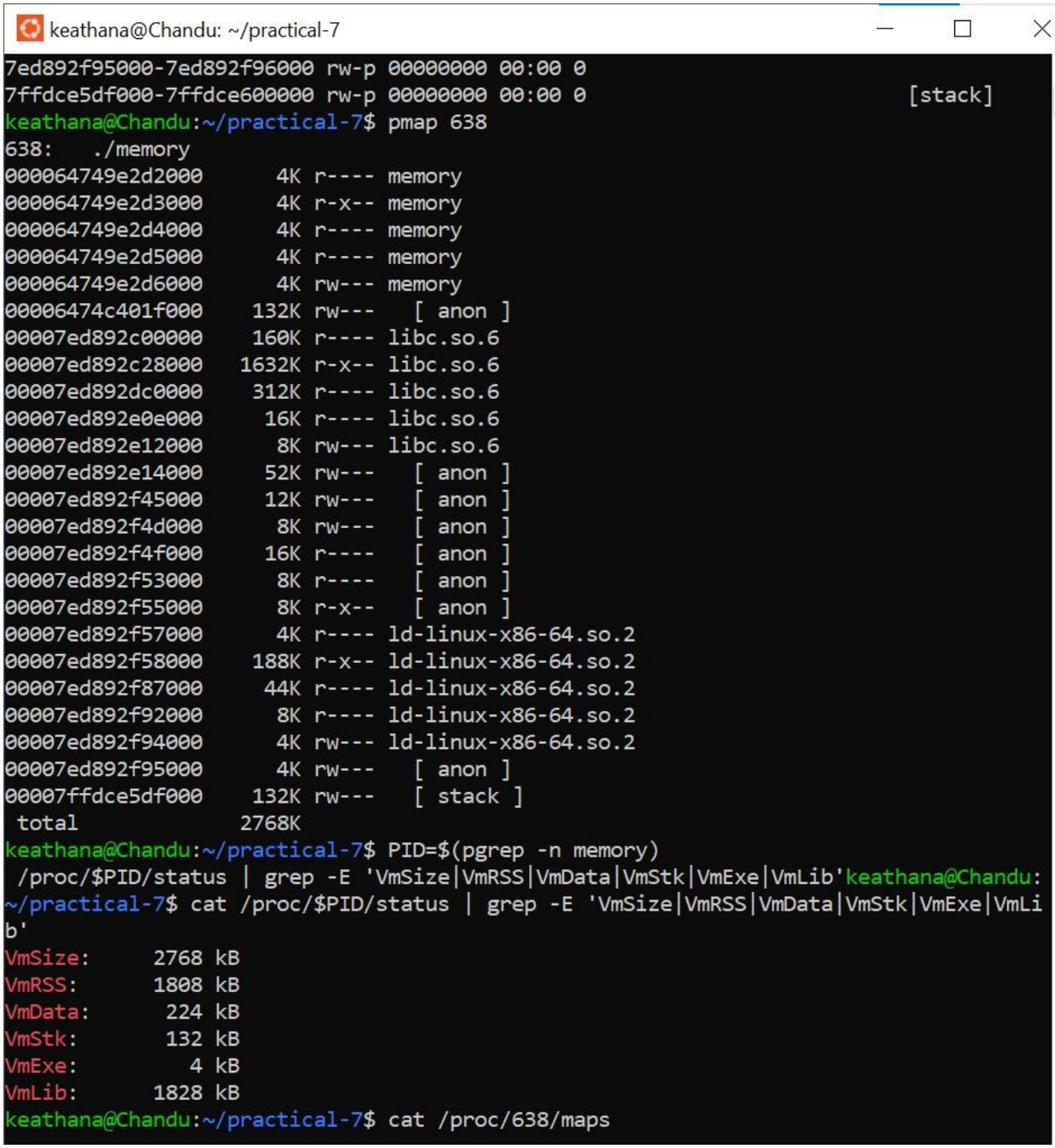
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

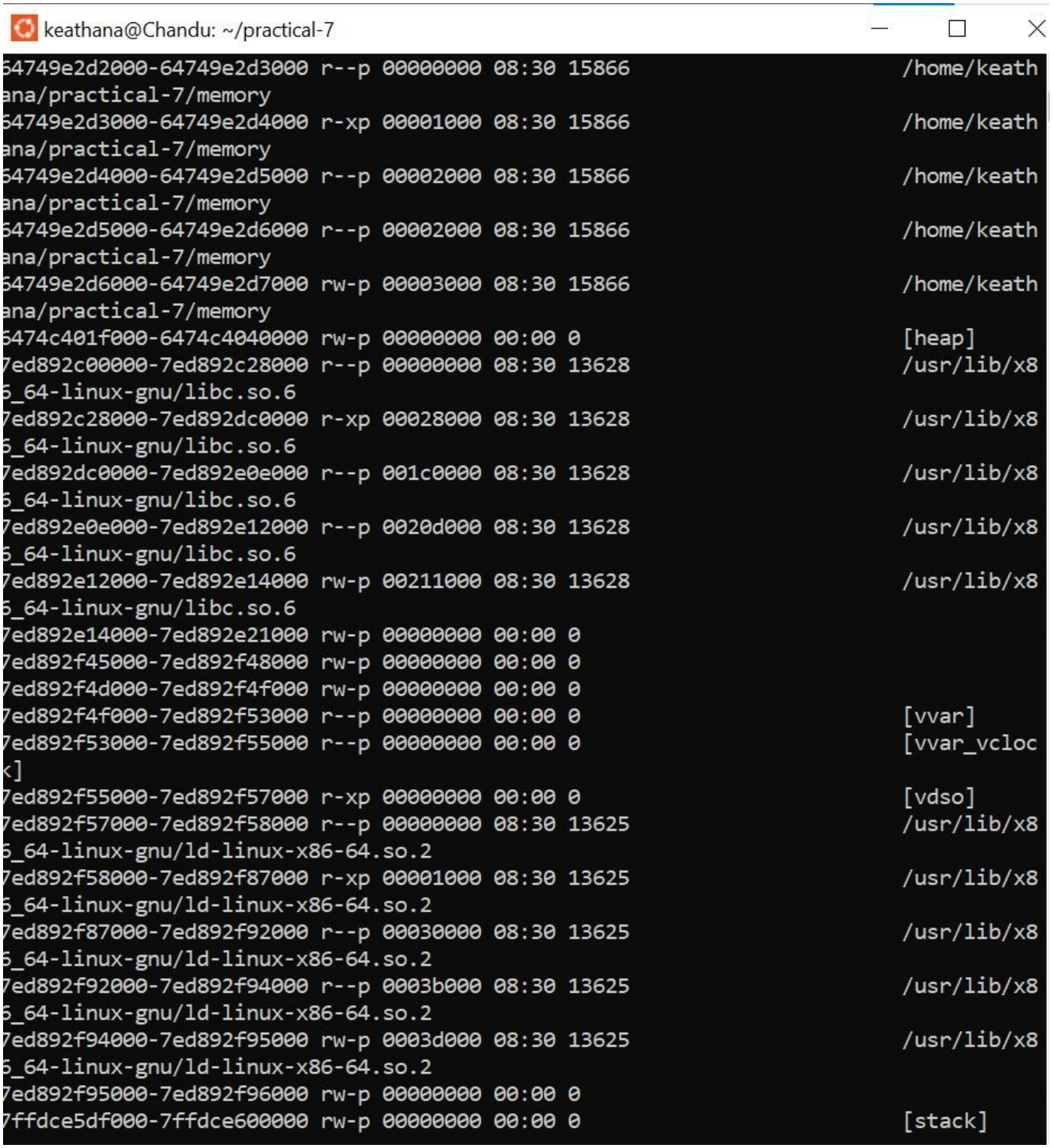

keathana@Chandu: ~/practical-7
GNU nano 8.7.1 memory.c *
int *heap_var = (int *)malloc(sizeof(int));
*heap_var = 40;
printf("==\n");
printf(" PRACTICAL 7 - MEMORY LAYOUT\n");
printf("==\n");
printf("Process ID : %d\n", getpid());
code_function();
printf("Address of global variable: %p\n",
(void *)&global_var);
printf("Address of static variable: %p\n",
(void *)&static_var);
printf("Address of heap variable : %p\n",
(void *)heap_var);
printf("Address of stack variable : %p\n",
(void *)&stack_var);
printf("\nProcess is running...\n");
printf("Use another terminal to inspect /proc/%d/maps\n",
getpid());
printf("Press Enter to terminate...\n");
getchar();
free(heap_var);
return 0;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

Output-


keathana@Chandu: ~/practical-7
keathana@Chandu:~/practical-7$ gcc memory.c -o memory
keathana@Chandu:~/practical-7$ ls
memory memory.c
keathana@Chandu:~/practical-7$ ./memory
=====================================
PRACTICAL 7 - MEMORY LAYOUT
=====================================
Process ID : 638
Address of code function : 0x64749e2d3209
Address of global variable: 0x64749e2d6010
Address of static variable: 0x64749e2d6014
Address of heap variable : 0x6474c401f010
Address of stack variable : 0x7ffdce5febac
Process is running...
Use another terminal to inspect /proc/638/maps
Press Enter to terminate...


keathana@Chandu: ~/practical-7
keathana@Chandu:~$ cd ~/practical-7
keathana@Chandu:~/practical-7$ cat /proc/638/maps
64749e2d2000-64749e2d3000 r--p 00000000 08:30 15866 /home/keathana/practical-7/memory
64749e2d3000-64749e2d4000 r-xp 00001000 08:30 15866 /home/keathana/practical-7/memory
64749e2d4000-64749e2d5000 r--p 00002000 08:30 15866 /home/keathana/practical-7/memory
64749e2d5000-64749e2d6000 r--p 00002000 08:30 15866 /home/keathana/practical-7/memory
64749e2d6000-64749e2d7000 rw-p 00003000 08:30 15866 /home/keathana/practical-7/memory
6474c401f000-6474c4040000 rw-p 00000000 00:00 0 [heap]
7ed892c00000-7ed892c28000 r--p 00000000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7ed892c28000-7ed892dc0000 r-xp 00028000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7ed892dc0000-7ed892e0e000 r--p 001c0000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7ed892e0e000-7ed892e12000 r--p 0020d000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7ed892e12000-7ed892e14000 rw-p 00211000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7ed892e14000-7ed892e21000 rw-p 00000000 00:00 0
7ed892f45000-7ed892f48000 rw-p 00000000 00:00 0
7ed892f4d000-7ed892f4f000 rw-p 00000000 00:00 0
7ed892f4f000-7ed892f53000 r--p 00000000 00:00 0 [vvar]
7ed892f53000-7ed892f55000 r--p 00000000 00:00 0 [vvar_vclock]
7ed892f55000-7ed892f57000 r-xp 00000000 00:00 0 [vdso]
7ed892f57000-7ed892f58000 r--p 00000000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7ed892f58000-7ed892f87000 r-xp 00001000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7ed892f87000-7ed892f92000 r--p 00030000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7ed892f92000-7ed892f94000 r--p 0003b000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7ed892f94000-7ed892f95000 rw-p 0003d000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2


keathana@Chandu: ~/practical-7
7ed892f95000-7ed892f96000 rw-p 00000000 00:00 0
7ffdce5df000-7ffdce600000 rw-p 00000000 00:00 0 [stack]
keathana@Chandu:~/practical-7$ pmap 638
638: ./memory
000064749e2d2000 4K r---- memory
000064749e2d3000 4K r-x-- memory
000064749e2d4000 4K r---- memory
000064749e2d5000 4K r---- memory
000064749e2d6000 4K rw--- memory
00006474c401f000 132K rw--- [anon]
00007ed892c00000 160K r---- libc.so.6
00007ed892c28000 1632K r-x-- libc.so.6
00007ed892dc0000 312K r---- libc.so.6
00007ed892e0e000 16K r---- libc.so.6
00007ed892e12000 8K rw--- libc.so.6
00007ed892e14000 52K rw--- [anon]
00007ed892f45000 12K rw--- [anon]
00007ed892f4d000 8K rw--- [anon]
00007ed892f4f000 16K r---- [anon]
00007ed892f53000 8K r---- [anon]
00007ed892f55000 8K r-x-- [anon]
00007ed892f57000 4K r---- ld-linux-x86-64.so.2
00007ed892f58000 188K r-x-- ld-linux-x86-64.so.2
00007ed892f87000 44K r---- ld-linux-x86-64.so.2
00007ed892f92000 8K r---- ld-linux-x86-64.so.2
00007ed892f94000 4K rw--- ld-linux-x86-64.so.2
00007ed892f95000 4K rw--- [anon]
00007ffdce5df000 132K rw--- [stack]
total 2768K
keathana@Chandu:~/practical-7$ PID=$(pgrep -n memory)
/proc/$PID/status | grep -E 'VmSize|VmRSS|VmData|VmStk|VmExe|VmLib'keathana@Chandu:
~/practical-7$ cat /proc/$PID/status | grep -E 'VmSize|VmRSS|VmData|VmStk|VmExe|VmLi
b'
VmSize: 2768 kB
VmRSS: 1808 kB
VmData: 224 kB
VmStk: 132 kB
VmExe: 4 kB
VmLib: 1828 kB
keathana@Chandu:~/practical-7$ cat /proc/638/maps


keathana@Chandu: ~/practical-7
64749e2d2000-64749e2d3000 r--p 00000000 08:30 15866 /home/keathana/practical-7/memory
64749e2d3000-64749e2d4000 r-xp 00001000 08:30 15866 /home/keathana/practical-7/memory
64749e2d4000-64749e2d5000 r--p 00002000 08:30 15866 /home/keathana/practical-7/memory
64749e2d5000-64749e2d6000 r--p 00002000 08:30 15866 /home/keathana/practical-7/memory
64749e2d6000-64749e2d7000 rw-p 00003000 08:30 15866 /home/keathana/practical-7/memory
6474c401f000-6474c4040000 rw-p 00000000 00:00 0 [heap]
7ed892c00000-7ed892c28000 r--p 00000000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7ed892c28000-7ed892dc0000 r-xp 00028000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7ed892dc0000-7ed892e0e000 r--p 001c0000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7ed892e0e000-7ed892e12000 r--p 0020d000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7ed892e12000-7ed892e14000 rw-p 00211000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7ed892e14000-7ed892e21000 rw-p 00000000 00:00 0
7ed892f45000-7ed892f48000 rw-p 00000000 00:00 0
7ed892f4d000-7ed892f4f000 rw-p 00000000 00:00 0
7ed892f4f000-7ed892f53000 r--p 00000000 00:00 0 [vvar]
7ed892f53000-7ed892f55000 r--p 00000000 00:00 0 [vvar_vclock]
7ed892f55000-7ed892f57000 r-xp 00000000 00:00 0 [vdso]
7ed892f57000-7ed892f58000 r--p 00000000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7ed892f58000-7ed892f87000 r-xp 00001000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7ed892f87000-7ed892f92000 r--p 00030000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7ed892f92000-7ed892f94000 r--p 0003b000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7ed892f94000-7ed892f95000 rw-p 0003d000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7ed892f95000-7ed892f96000 rw-p 00000000 00:00 0
7ffdce5df000-7ffdce600000 rw-p 00000000 00:00 0 [stack]